

Metal disc couplings | Arcoflex type 314

800 - 9000Nm



- Backlash-free and torsionally rigid
- Hubs mount externally or internally (style Y)
- Maintenance free
- Pre-assembled disc packs
- High speed capacity
- Resistant to chemicals and ambients from -40°C to +270°C



Selection options

Rated torques to achieve long life are based on the actual misalignment of the installation, 0.25° is best case, 1.3° is worst case. Lower figures apply if reversing.

Hubs

Standard hubs to style X have larger bore capacity but will not fit inside the spacers. To achieve this use one or two hubs of style Y.

Spacers

Two spacer lengths are available, long to Design 2.1 with length L2 and short to Design 4.1 with length L8. Design 4.1 is more compact but has lower misalignment capacity.

Size	Torque at angle of misalignment per disc pack Nm				Bores d, d1			Max speed r/min	Torsional stiffness Nm/°		Inertia kgm ² x 10 ⁻³			Weight kg		
	0.5°	1°	1.3°	Reversing†	Min	Max	Max		1.1	2/4.1	1.1	2.1	4.1	1.1	2.1	4.1
55	800	600	400	250	25	65	55	6700	6982	3491	6.1	10.2	9.8	4.2	5.7	5.5
65	1200	900	600	300	30	75	65	5900	14138	6982	11.8	18.7	17.7	6.4	8.5	8.1
75	1800	1400	1000	550	35	85	75	5100	12742	6458	23.8	37.5	35.6	9.6	12.5	12.1
80	2800	2000	1500	850	35	90	80	4750	16058	8029	36.5	59	56	12.5	16.5	16.2
85	4500	2500	—	1600	40	100	85	4300	22866	11520	57	95	91	15.5	21	20.5
90	6000	3000	—	1900	45	110	90	4000	52016	26008	83	138	133	19.5	27	26
98	9000	4000	—	3500	50	120	100	3400	104555	52190	174	294	285	30	42	40.6

† Use this rating where either direction or torque loads are reversing.

Maximum misalignments

Size	Angular		Axial		Radial		
	1.1	2.1 & 4.1	1.1	2.1 & 4.1	1.1	2.1	4.1
55	1.3°	2.6°	1.0	2.0	NOT PERMITTED	1.75	1.0
65	1.3°	2.6°	1.1	2.2		2.15	1.2
75	1.3°	2.6°	1.3	2.6		2.5	1.4
80	1.3°	2.6°	1.3	2.6		2.55	1.5
85	1°	2°	1.0	2.0		2.0	1.1
90	1°	2°	1.2	2.4		2.5	1.4
98	1°	2°	1.4	2.8		2.0	1.5

Ordering example

8 off Arcoflex couplings model 314.65.2.1 with style X hubs, d1 = pilot bore, d2 = 65H7, with Design 2:1 central spacer assembly.

8 off Type No. 13010940

8 off Type No. 13011010

8 off Type No. 13024375

Type numbers, clamp hubs and spacer couplings, see following pages.

Materials

Hubs and centre spacers – Carbon steel
Disc pack – Stainless steel

Temperature

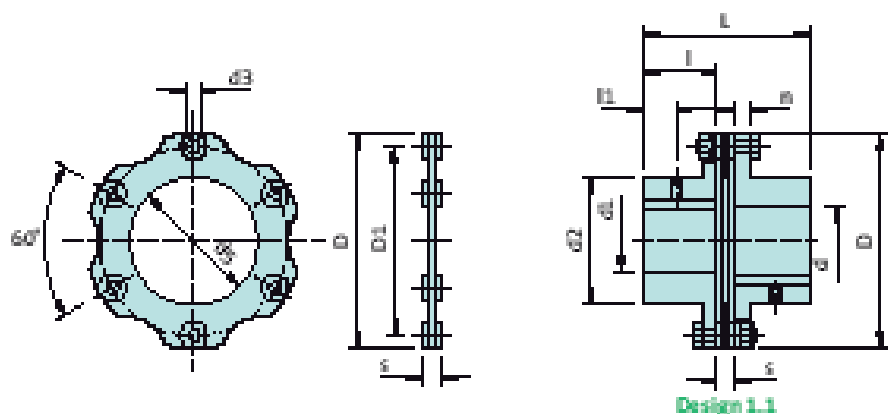
Ambient temperatures up to 270°C are possible but performance data must be derated above 150°C. Ask for details.

Fixing bolts

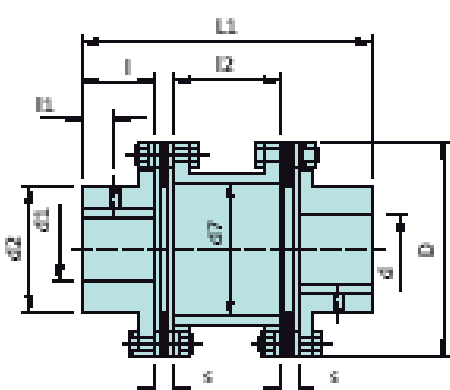
High tensile fitted hexagonal head bolts are supplied with all metal locknuts and washers. The minimum clearance to remove bolts is given by dimension m.

Size	k	m(min)	Screw tightening torque Nm	
55	5.5	38	M8	33
65	5.5	38	M8	33
75	7	46	M10	65
80	8	54	M12	115
85	10	61	M16	280
90	10	66	M16	280
98	13	79	M20	550

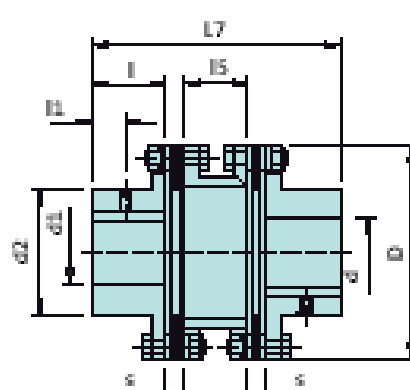
Bolts should be fitted as received. If re-lubrication is required, only light oil should be used.



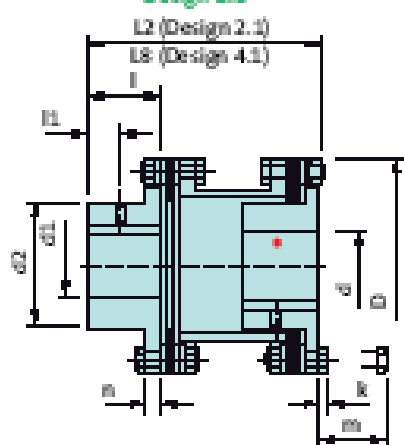
Design 1.1



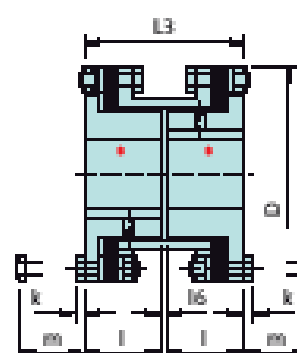
Design 2.1



Design 4.1



Design 2.1/4.1 (1 hub style Y)



Design 2.1 (2 hubs style Y)

Size	D	Ds	ds	d 2'	ds	ds	ds	L	Ls	Ls	La	La	La	l	ls	ls	ls	ls	n	s	m
							H7														
55	128	112	88	77	8	80	77	121	206	160	114	170	124	55	20	74	38	4	9	11	38
65	145	128	103	90	8	94	90	141	246	190	134	200	144	65	25	94	48	4	9	11	38
75	168	148	117	104	11	108	104	164	286	220	154	233	167	75	30	108	55	4	9	14	46
80	180	158	125	112	13	116	112	175	300	232	164	246	179	80	30	110	56	4	12	15	59
85	200	170	134	119	17	124	119	175	300	233	165	246	179	80	30	110	56	4	13	15	61
90	215	185	147	128	17	130	128	200	340	263	186	281	204	90	35	120	61	4	13	20	66
98	250	214	164	145	21	149	145	223	370	288	205	309	227	100	40	134	63	4	18	23	79

* Style Y hubs only for internal mounting

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Size	Hubs								Central assemblies	
	Style X						Style Y			
	Bore	Type No.	Bore	Type No.	Bore	Type No.	Bore	Type No.	Design	Type No.
55	25P8	13010935	38H7	13010614	50H7	13011004	25P8	13010850	1.1	13024398
	32H7	13010613	60H7	13010612	55H7	13011005	50H7	13024368	2.1	13024374
	35H7	13011002	65H7	13011003	65H7	13011006	55H7	-	4.1	13024443
65	30P8	13010940	60H7	13010616	55H7	13011009	30P8	13010858	1.1	13024455
	32H7	13032703	65H7	13011007	65H7	13011010	55H7	-	2.1	13024375
	38H7	13010615	50H7	13011008	75H7	13011011	65H7	-	4.1	13024454
75	35P8	13010929	55H7	13009805	70H7	13003620	35P8	13010851	1.1	13024456
	68H7	13010617	60H7	13009806	80H7	13003632	60H7	-	2.1	13024376
	50H7	13011012	65H7	13003622	85H7	13009807	75H7	-	4.1	13024457
80	35P8	13009808	55H7	13003628	70H7	13009811	35P8	-	1.1	13024458
	68H7	-	60H7	13009810	80H7	13010602	60H7	-	2.1	13024377
	50H7	13009809	65H7	13003641	90H7	13010603	80H7	-	4.1	13024463
85	60P8	13010943	65H7	-	80H7	-	60P8	13010859	1.1	13024459
	55H7	-	70H7	-	90H7	-	70H7	-	2.1	13024378
	60H7	-	75H7	-	1.00H7	-	90H7	-	4.1	13024462
90	65P8	13010944	75H7	-	1.00H7	-	65P8	13010855	1.1	13024460
	60H7	-	80H7	-	1.05H7	-	80H7	-	2.1	13024379
	70H7	-	90H7	-	1.10H7	-	90H7	-	4.1	13024464
98	50P8	13010945	80H7	-	1.05H7	-	50P8	13010852	1.1	13024461
	70H7	-	90H7	-	1.10H7	-	90H7	-	2.1	-
	75H7	-	1.00H7	13010611	1.20H7	-	100H7	-	4.1	13024465

Type no. in blue are usually available from stock. Where no type no. is shown, enquire for a price and delivery.

Keyways to BS4235

Other bore sizes available please ask.